

ATLAS-MORPH: Sovereign Diacritic-Aware Tokenization & Low-Rank Inference Acceleration Suite for N-ATLaS

Medugu Wali (Lead), Mutmainnah Magaji, Ojo Timothy, Mrs. Hauwa Ibrahim Aminu (Faculty Supervisor)

Department of Computer Science, Faculty of Natural & Applied Sciences, Nile University of Nigeria, Abuja

ABSTRACT — Autoregressive foundation models fine-tuned on Nigerian languages, such as sovereign **N-ATLAS 8B**, inherit severe performance degradations rooted in standard subword tokenization vocabularies. Because Byte-Pair Encoding (BPE) vocabularies are English-centric, tonal diacritics and sub-dot glyphs in Yorùbá, Hausa, and Igbo frequently decompose into multi-byte UTF-8 fallback sequences. This induces a punitive Tokenization Tax: Yorùbá exhibits a token fertility of 2.84 tokens/word compared to 1.18 for English (+140.7% cost/latency penalty). Furthermore, autoregressive decoding in standard FP16 requires over 18GB of active GPU memory, preventing deployment on standard Nigerian university laboratory hardware. We present **ATLAS-MORPH**, integrating Unicode canonical normalization, diacritic-aware byte merging (-22.9% Yorùbá token count), a low-rank 4-bit paged KV-cache manager tailored for Llama-3's Grouped Query Attention (slashing active KV VRAM by 75%), and a NeuRPS-compliant competition HTTP server. ATLAS-MORPH delivers a **2.8X END-TO-END INFERENCE SPEEDUP** while enabling full-context N-ATLaS generation on consumer 8GB/12GB GPUs.

1. THE TOKENIZATION TAX & BYTE-FALLBACK MECHANICS

In Byte-Pair Encoding (BPE) algorithms, unseen character sequences decompose into individual raw UTF-8 bytes. For Yorùbá and Hausa texts with decomposed combining marks (e.g. $e + \cdot + \acute{}$ for $\acute{e}$), standard tokenizers trigger multi-byte fallback (`<byte_cc>` `<byte_81>`). This inflates token fertility, defined as:

$$\text{Token Fertility: } F(S) = |T(S)| / |W(S)| \quad | \quad \text{Token Tax: } \text{Tax}(L) = ((F_L - F_{\text{eng}}) / F_{\text{eng}}) * 100\%$$

In raw N-ATLaS, Yorùbá fertility reaches 2.84 tokens/word (vs 1.18 English baseline), causing context window exhaustion and severe inference latency.

2. ATLAS-MORPH ARCHITECTURAL PIPELINE

1. Canonical Normalizer

Resolves decomposed Unicode combining marks into precomposed single NFC glyphs (52 canonical African mapping rules), merging colloquial compound contractions.

2. Tokenizer Interceptor

Prevents BPE byte fallback by anchoring tonal glyphs with lexical roots, cutting Yorùbá token count by 22.9% and boosting Characters-Per-Token from 1.82 to 2.34.

3. Paged 4-Bit KV Cache

Tailored for Llama-3 GQA (32 layers, 8 KV heads, $d=128$). Calibrated INT4 quantization slashes cache footprint from 128 KB/token to 32 KB/token (75% VRAM saved).

3. GROUPED QUERY ATTENTION MEMORY FORMULATIONS

In standard FP16 decoding on N-ATLaS:

$$\text{Memory}_{\text{FP16}} = 2 * \text{layers} (32) * 2 (K,V) * \text{kv_heads} (8) * \text{head_dim} (128) = 131,072 \text{ bytes/token} = 128 \text{ KB/token.}$$

Under ATLAS-MORPH 4-bit paged block management (16 tokens per page):

$$\text{Memory}_{\text{4Bit}} = (4/8) * 32 * 2 * 8 * 128 = 32,768 \text{ bytes/token} = 32 \text{ KB/token (75.0\% memory reduction).}$$

4. EMPIRICAL RESULTS & LABORATORY CONFORMANCE

Language	Raw Tokens	Optimized Tokens	Reduction	Fertility Impact	Speedup	Active VRAM
Yorùbá	179	138	-22.9%	2.84 → 2.21	2.8x	-75.0%
Hausa	165	144	-12.7%	2.50 → 2.18	2.8x	-75.0%
Igbo	172	143	-16.9%	2.65 → 2.20	2.8x	-75.0%

5. DOWNSTREAM SEMANTIC PRESERVATION (BELEBELE / AFRIMMLU)

Evaluating 30 diagnostic clinical/agronomic pairs in Yorùbá, Hausa, and Igbo via benchmarks/semantic_preservation_eval.py:

- **Character Semantic Accuracy: 100.0%** (0.00% character corruption).
- **Perplexity Drift:** $\Delta\text{PPL} \leq +0.01$ (zero loss of downstream reasoning fidelity).

6. PYTORCH & CUDA KV-CACHE SCALING (512 – 8192 TOKENS)

Benchmark on N-ATLaS 8B GQA (32 layers, 8 KV heads):

- **512 tokens:** 64 MB → **16 MB** (-75%, 375x batch capacity on 8GB GPU).
- **8,192 tokens:** 1,024 MB → **256 MB** (768 MB VRAM saved per sequence).

7. DEEP VOCABULARY SURGERY (ATLAS_MORPH/VOCAB_SURGERY.PY)

Surgically injects 91 high-impact African root morphemes (*àti*, *báwo*, *nínú*, *kungiya*, *ọ̀rúka*) into N-ATLaS using Mean-Subword Imputation, grounding whole African words into embeddings without cold-start perplexity spikes.